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Professor

Department of Ecology and Evolutionary Biology
University of Arizona, Tucson, Arizona, USA

External Professor

The Santa Fe Institute,
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Oxford University
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Leverhulme Centre
Oxford, UK

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Bridging Biodiversity and Conservation Science
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Education

- PhD Biology (Ecology Program) July 1998. University New Mexico, Albuquerque, NM, USA; Major Advisor, James H. Brown
- MS Biology (Ecology Program) Feb.1994. University of New Mexico, Albuquerque, NM, USA
- BA Biology, (*With Distinction*), May 1991. Colorado College Colorado Springs, CO, USA

Professional Employment Appointments

- **External Faculty, The Santa Fe Institute**, Santa Fe, New Mexico (2007-).
- **Professor, Department of Ecology & Evolutionary Biology, The University of Arizona**, Tucson, Arizona (2009 -).
- **Honorary Research Associate, Oxford University, Oxford, UK (2025-)**
- Associate Professor, Department of Ecology & Evolutionary Biology, The University of Arizona, Tucson, Arizona (2005- 2009).
- Assistant Professor, Department of Ecology Evolutionary Biology, The University of Arizona, Tucson, Arizona (2001-2005).
- NSF Postdoctoral Fellow, NCEAS (National Center for Ecological Analysis and Synthesis), University of California, Santa Barbara, CA (Sept. 1999 – Dec. 2000).

- NSF Postdoctoral Fellow, SFI (The Santa Fe Institute), Santa Fe, New Mexico. (Aug. 1998 - Sept 1999).

Academic Honors

- Ecological Society of America, ESA Fellow (2018)
- American Association for the Advancement of Science, AAAS Fellow (2012).
- Honorary, Doctor of Science – D.Sci, (Honoris Causa) The Colorado College (2007).

Awards, Fellowships, and Honors

Fulbright Fellow, National Geographic Explorer, Oxford University Martin School Fellow, Web of Science Highly Cited Researcher, ESA George C. Mercer Award, ESA MacArthur Award, National Science Foundation CAREER Award

- [MacArthur Award, Ecological Society of America, \(2026\)](#).
- National Academy of Sciences panel on developing a national report on [Continental Scale Biology](#) (2023-2024).
- [Web of Science, Highly Cited Researcher](#) (2018, 2019, 2020, 2025)
- [National Geographic Explorer](#) (2020)
- [Leverhulme Professorship](#), Oxford University, UK (2017).
- Martin School Fellow, Oxford University, UK (2017).
- College of Science Galileo Circle Fellow, University of Arizona (2011).
- Elected Chair, Gordon Research Conference, Metabolic Basis Ecology Evolution (2010).
- Charles University, Center for Theoretical Study, International Mobility Fellow, Prague, Czech Republic (2009).
- Centre National de la Recherche Scientifique (CNRS), Associate Research Fellow, Montpellier, France (2009).
- TEAM Fellow, Conservation International (2007).
- ISI Essential Science Indicators (ESI), Author with the highest percent increase in total citations in the field of Environment and Ecology (2005).
- Popular Science Magazine, “Top 10 Brilliant Young Scientists” (2004).
- NSF CAREER Young Investigators Award (2002-2007).
- Center for Applied Biodiversity Science Fellow, Conservation International (2002-2004)
- George C. Mercer Award, Ecological Society of America, *Awarded for best ecological paper by a researcher under the age of 40* (2001).
- NSF Postdoctoral Research Fellowship (1998-2000).
- Santa Fe Institute Postdoctoral Research Fellowship, *Declined* (1998).
- Visiting Scholar, Silwood Park, Imperial College, UK (May, 1998).
- Best research poster, UNM Biology Research Day (1994 & 1998).
- NSF Graduate Research Training Fellowship (1996-1998).
- Santa Fe Institute Complex Systems Summer School (1997).
- Fulbright Fellow, Costa Rica (1995-1996).

- Richard G. and Reba Beidleman Award, Colorado College, Honors Convocation, *For Excellence in Ecology* (1991).
- Graduation with Distinction in Biology, Colorado College (1991).

Editorial Boards

- Associate Editor, Ecology Letters (Nov. 2007 – 2025).
- Associate Editor, Global Ecology and Biogeography (Summer 2010 - 2025).
- Invited Guest Editor, Annual Review of Ecology, Evolution, & Systematics (2009 - 2010).
- Editorial Board, Global Ecology and Biogeography (2003- 2006).
- Member of Faculty of 1000 (Nov. 2005 – Jan. 2007).

Social Media/Open Science

- Bluesky - <https://bsky.app/profile/bjenquist.bsky.social>
- GitHub - <https://github.com/EnquistLab>
- Enquist lab - [YouTube Channel](#)
- FigShare - https://figshare.com/authors/Brian_Enquist/663712
- Lab website and blog - <https://brianjenquist.wordpress.com>
- Google Scholar - <https://scholar.google.com/citations?user=mAbA6EoAAAAJ&hl=en>
- ResearchGate - https://www.researchgate.net/profile/Brian_Enquist

Scholarly Publications

- Google Scholar, 05/13/2026: Citations = 74,123, h-index = 118.

Books

- Enquist, B.J., Kempes, C., O'Connor, M. ***Biological Scaling: A general theory for scaling the form and function of biology and biodiversity***. Santa Fe Institute Press (In Prep).
- National Academies of Sciences, Engineering, and Medicine. 2024. ***A Vision for Continental-Scale Biology: Research Across Multiple Scales***. Washington, DC: The National Academies Press. <https://doi.org/10.17226/27285>.

Reports

- **UNESCO and IUCN (2023) UNESCO - World Heritage: A unique contribution to biodiversity conservation.** by the United Nations Educational, Scientific and Cultural Organization (UNESCO), 7, place de Fontenoy, 75352 Paris 07 SP, France, and the International Union for Conservation of Nature (IUCN), Rue Mauverney 28, 1196 Gland, Switzerland. <https://unesdoc.unesco.org/ark:/48223/pf0000385392>
https://sparc-apps.shinyapps.io/biodiversity_explorer/

- **IPBES (Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services) Chapter 3 Status and trends of biodiversity and ecosystem functions underpinning nature's benefit to people.** In IPBES (2018): The IPBES regional assessment report on biodiversity and ecosystem services for the Americas. Rice, J., Seixas, C. S., Zaccagnini, M. E., Bedoya-Gaitán, M., and Valderrama, N. (eds.). Secretariat of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services, Bonn, Germany, pp. 207-297.
<https://www.ipbes.net/system/tdf/ipbes-6-inf-4-rev.1.pdf?file=1&type=node&id=16517>

Peer Reviewed Journal Aricles

2026

1. Enquist, B.J. and Kempes, C.P. (2026) From Cells to the Biosphere: The Expanding Role of Metabolic Scaling Theory in Biology. *Annual Review of Ecology, Evolution, and Systematics*. (In press).
2. Song, L., Frazier, A. E., Kedron, P., Araujo, D. S. A., Cui, D., Enquist, B. J., Hinojo-Hinojo, C., Krieger, J. M., Maitner, B. S., Merow, C., Moulatlet, G. M., Nikolopoulos, E. I., & Roehrdanz, P. R. (in press). Explainable artificial intelligence reveals spatially divergent effects of global change on mammals. *Nature Communications* (In Press)
3. Michiel D. Pillet*, Brian S. Maitner, Mónica Arakaki, Diogo S. A. Araujo, Marc A. Baker, Duniel Barrios, Bradley L. Boyle, David D. Breshears, Bárbara Goettsch, Xiao Feng, Pablo C. Guerrero, Tania Hernández-Hernández, Lucas C. Majure, Beatriz M. Meriño, Efthymios I. Nikolopoulos, Raúl Puente-Martínez, Josep M. Serra-Diaz, Patrick R. Roehrdanz, Cory Merow, Brian J. Enquist. Accounting for severe droughts increases the extinction risk for cacti. (in press) *Nature*
4. Enquist, B.J. et al. (2026) BIEN: [A biodiversity informatics ecosystem advancing open and reproducible workflows for plant observation, plot, and trait data](#). *Methods in Ecology and Evolution*. 00, 1–29.
5. Navarro, J., Cruz-de Hoyos, R.M., Powers, J.M., Blonder, B.W. and Enquist, B.J., 2026. [Ploidy level predicts differences in minimum leaf conductance in quaking aspen](#), *Populus tremuloides*. *American Journal of Botany*, p.e70235.
6. Althuizen, I.H., Gya, R., Jaroszynska, F., Lee, H., Telford, R.J., Chipperfield, J., Enquist, B.J., Goldberg, D.E. and Vandvik, V., 2026. [Climate-induced shifts in plant investment strategies regulate ecosystem carbon cycling across alpine grasslands](#). *Journal of Ecology*, 114(6), p.e70364.
7. Guo, W.-Y., Serra-Diaz, J. M., Guo, K., Boonman, C. C. F., Schrod, F., Maitner, B. S., Merow, C., Violle, C., Anand, M., Bruun, H. H. K., Byun, C., Catford, J. A., Cerabolini, B. E. L., Chacón-Madrigal, E., Ciccarelli, D., Dang-Le, A. T., Dias, A. S., Giroldo, A. B., Gutiérrez, Á. G., Jansen, S., Kattge, J., Kindt, R., Klein, T., Kramer, K., Lusk, C. H., Martin, A. R., Michaletz, S. T., Minden, V., Mori, A. S., Niinemets, Ü., Onoda, Y., Peñuelas, J., Pisek, J., Robroek, B. J. M., Schamp, B., Soudzilovskaia, N. A., Thiffault, N., van der Plas, F., Enquist, B. J., & Svenning, J.-C. (2026). [Global functional shifts in trees driven by alien naturalization and native extinction](#). *Nature Plants* In Press.

8. Brock, C., Roehrdanz, P. R., Beringer, T., Chaplin-Kramer, R., Enquist, B. J., Frazier, A. E., Johnson, J. A., Kennedy, C. M., Kiesecker, J., Larsen, A. E., Loyola, R., Marquet, P. A., Neugarten, R. A., Oakleaf, J. R., Roopsind, A., Schuster, R., Williams, D. R., Wu, G. C., Zvoleff, A., & Hannah, L. (2026). [Balancing land use for conservation, agriculture, and renewable energy](#). *Nature Communications*, 17, 3623
<https://doi.org/10.1038/s41467-026-69952-6>

2025

9. Moulatlet, G. M., Merow, C., Maitner, B., Boyle, B., Feng, X., Frazier, A. E., Hinojo-Hinojo, C., Newman, E. A., Roehrdanz, P. R., Song, L., Villalobos, F., Marquet, P. A., Svenning, J.-C., & Enquist, B. J. (2025). [General laws of biodiversity: Climatic niches predict plant range size and ecological dominance globally](#). *Proceedings of the National Academy of Sciences* 122.46: e2517585122.
10. Zuidema, Pieter A., et al. [Pantropical tree rings show small effects of drought on stem growth](#). *Science* 389.6759 (2025): 532-538.
11. Vasseur, F., Mahaut, L., Enquist, B.J., and C. Violle (2025) [From organism traits to ecosystem processes: why size is so important](#). *Annual Review of Ecology, Evolution, Systematics* 56.
12. Boonman, C.C., Hoeks, S., Serra-Diaz, J.M., Guo, W.Y., Enquist, B.J., Maitner, B., Merow, C. and Svenning, J.C., 2025. [High tree diversity exposed to unprecedented macroclimatic conditions even under minimal anthropogenic climate change](#). *Proceedings of the National Academy of Sciences*, 122(26), p.e2420059122.
13. Cruz, A. R., & Enquist, B. J. (2025). [Allometry and phylogeny of within-diaspore biomass allocation: A global analysis](#). *Biology Letters*, 21(11).
14. Mouquet, N., Mahaut, L., Thuiller, W., Auber, A., Casajus, N., Enquist, B. J., ... & Violle, C. (2025). [Spatial Insurance of Distinct Ecological Functions](#). *Ecology Letters*, 28(11), e70266.
15. Halbritter, Aud H., et al. Effects of Warming, [Nitrogen and Grazing on Plant Functional Traits Differ Between Alpine and Sub-Alpine Grasslands](#). *Journal of Vegetation Science* 36.5 (2025): e70061.
16. Gaudard, J., Telford, R. J., Chacon-Labela, J., Dawson, H. R., Enquist, B. J., Töpper, J. P., ... & Halbritter, A. H. (2025). [flexible: An R package to process ecosystem gas fluxes from closed-loop chambers in an automated and reproducible way](#). *Methods in Ecology and Evolution*.
17. Feng X, Smith AB, Boyle B, Chen X, Enquist BJ, Gallagher R, Hammock J, Ho JC, Lien AM, Maitner B, Sokol ER. (2025). [The next stage of biodiversity informatics: community-driven synthesis and integration of biodiversity databases](#). *BioScience*, 75(11), 913-925.
18. Halbritter, A. H., Vandvik, V., Bison, N. N., Clark, V. R., Cross, M., Greve, M., ... & Enquist, B. J. (2025). [Plant traits and associated ecological data from Afromontane grasslands of Maloti-Drakensberg, South Africa](#). *Scientific data*, 12(1), 1778.
19. Vandvik, V., Halbritter, A. H., Macias-Fauria, M., Maitner, B. S., Michaletz, S. T., Telford, R. J., ... & Enquist, B. J. (2025). [Plant traits and associated ecological data from global change experiments and climate gradients in Norway](#). *Scientific data*, 12(1), 1477.
20. Hordijk, I., Poorter, L., Liang, J., Reich, P. B., De-Miguel, S., Nabuurs, G. J., et al. & Pfautsch, S. (2025). [Effect of climate on traits of dominant and rare tree species in the world's forests](#). *Nature Communications*, 16(1), 4773.

21. Groenendijk, P., Babst, F., Trouet, V., Fan, Z. X., Granato-Souza, D., Locosselli, G. M., ... & Wils, T. (2025). [The importance of tropical tree-ring chronologies for global change research](#). *Quaternary Science Reviews*, 355, 109233.
22. Veldhuisen, L. N., Zepeda, V., Enquist, B. J., & Dlugosch, K. M. (2025). [Rare species do not disproportionately contribute to phylogenetic diversity in a subalpine plant community](#). *American Journal of Botany*, e70061.
23. Matos, Ilaine Silveira, et al. [Leaf venation network evolution across clades and scales](#). *Nature Plants* (2025): 1-15.
24. Roos, Ruben E., et al. [Marine-derived nutrients shape the functional composition of High Arctic plant communities](#). *Functional Ecology* (2025).
25. Aguirre-Gutiérrez, Jesús, et al. [Tropical forests in the Americas are changing too slowly to track climate change](#). *Science* 387.6738 (2025): eadl5414.
26. Aguirre-Gutiérrez, J., Rifai, S.W., Deng, X. et al. [Canopy functional trait variation across Earth's tropical forests](#). *Nature* (2025).
27. Cruz, A. R., Enquist, B. J., & Burger, J. R. (2025). [Scaling COVID-19 rates with population size in the United States](#). *Journal of the Royal Society Interface*, 22(224), 20240839.
28. Lowe, A.J., Royer, D.L., Wiczyński, D.J., Butrim, M.J., Reichgelt, T., Azevedo-Schmidt, L., Peppe, D.J., Enquist, B.J., Kerkoff, A.J., Michaletz, S.T. and Strömberg, C.A., 2025. [Global patterns in community-scale leaf mass per area distributions of extant woody non-monocot angiosperms and their utility in the fossil record](#). *American Journal of Botany*, 112(4), p.e70019.
29. Araujo, D.S., Enquist, B.J., Frazier, A.E., Merow, C., Roehrdanz, P.R., Moulatlet, G.M., Zvoleff, A., Song, L., Maitner, B. and Nikolopoulos, E.I., 2025. Global Future Drought Layers Based on Downscaled CMIP6 Models and Multiple Socioeconomic Pathways. *Scientific Data*, 12(1), p.295. [Global Future Drought Layers Based on Downscaled CMIP6 Models and Multiple Socioeconomic Pathways](#). *Scientific Data* 12, no. 1 (2025): 295.

2024

30. Enquist, B.J., Kempes, C.P. and G.B. West (2024) [Developing a predictive Science of the Biosphere requires the integration of scientific cultures](#). *Proceedings of the National Academy of Sciences*, 121(19), e2209196121. <https://www.pnas.org/doi/10.1073/pnas.2209196121>
31. Westgeest, Adrianus J., François Vasseur, Brian J. Enquist, Rubén Milla, Alicia Gómez-Fernández, David Pot, Denis Vile, and Cyrille Violle. [An allometry perspective on crops](#). *New Phytologist* 244, no. 4 (2024): 1223-1237.
32. Enquist, B.J., Erwin, D., Savage, V., and P.A. Marquet (2024) [Scaling approaches and Macroecology provide a foundation for assessing ecological resilience in the Anthropocene](#). *Philosophical Transactions B*, 379: 20230010.
33. Bektaş, Billur, et al. [Colonization and extinction lags drive non-linear responses to warming in mountain plant communities across the Northern Hemisphere](#). *Ecography* (2024): e07378.
34. Mo, Lidong, et al. [The global distribution and drivers of wood density and their impact on forest carbon stocks](#). *Nature Ecology & Evolution* (2024): 1-18.
35. Grady, J.M., Read, Q.D., Record, S., Rüger, N., Zarnetske, P.L., Dell, A.I., Hubbell, S.P., Michaletz, S.T. and B.J. Enquist (2024) [Life history scaling in a tropical forest](#). *Journal of Ecology*. 112(3), pp.487-500.

36. Hordijk, Iris, et al. [Dominance and rarity in tree communities across the globe: Patterns, predictors and threats](#). *Global Ecology and Biogeography* 33.10 (2024): e13889.
37. Boonman, CCF, J.M. Serra-Diaz, S.Hoeks, W. Guo, B.J. Enquist, B. Maitner, Y.Malhi, C. Merow, R. Buitenwerf, and J.Svenning (2024) [More than 17,000 tree species are at risk from rapid global change](#). *Nature Communications* 15, no. 1: 166.
38. Maitner, Brian, et al. [Code sharing in ecology and evolution increases citation rates but remains uncommon](#). *Ecology and Evolution* 14.8 (2024): e70030.
39. Halbritter, A. H., Vandvik, V., Cotner, S. H., Farfan-Rios, W., Maitner, B. S., Michaletz, S. T., ... & Enquist, B. J. (2024). [Plant trait and vegetation data along a 1314 m elevation gradient with fire history in Puna grasslands, Perú](#). *Scientific Data*, 11(1), 225.
40. Ngute, Alain Senghor K., et al. (2024) [Global dominance of lianas over trees is driven by forest disturbance, climate and topography](#). *Global Change Biology* 30.1: e17140.
41. Martínez-Villa, J. A., Durán, S. M., Enquist, B. J., Duque, A., Messier, C., & Paquette, A. (2024). [Temporal shifts in the functional composition of Andean forests at different elevations are driven by climate change](#). *Global Ecology and Biogeography*, 33(1), 85-99.
42. Serra-Diaz JM, Borderieux J, Maitner B, Boonman CC, Park D, Guo WY, Callebaut A, Enquist BJ, Svenning JC, Merow C. [occTest: An integrated approach for quality control of species occurrence data](#). *Global Ecology and Biogeography*. 2024:e13847

2023

43. Maitner, B. S., Halbritter, A. H., Telford, R. J., Strydom, T., Chacon, J., Lamanna, C., ... & Enquist, B. J. (2023). [Bootstrapping outperforms community-weighted approaches for estimating the shapes of phenotypic distributions](#). *Methods in Ecology and Evolution*, 14(10), 2592-2610.
44. Mo, Lidong, et al. (2023) [Integrated global assessment of the natural forest carbon potential](#). *Nature*: 1-10.
45. Ma, Haozhi, et al. (2023) [The global biogeography of tree leaf form and habit](#). *Nature Plants* 9.11: 1795-1809.
46. Gallagher, Rachael V., et al. [Global shortfalls in threat assessments for endemic flora by country](#). *Plants, People, Planet* 5.6 (2023): 885-898.
47. Guo, Wen-Yong, et al. [Climate change and land use threaten global hotspots of phylogenetic endemism for trees](#). *Nature Communications* 14.1 (2023): 6950.
48. Vandvik, V., Halbritter, A. H., Althuizen, I. H., Christiansen, C. T., Henn, J. J., Jónsdóttir, I. S., ... & Enquist, B. J. (2023). [Plant traits and associated data from a warming experiment, a seabird colony, and along elevation in Svalbard](#). *Scientific Data*, 10(1), 578.
49. Delavaux, C. et al. (2023) [Native diversity buffers against severity of non-native tree invasions](#). *Nature* 621,773–781.
50. Duncanson L, Liang M, Leitold V, Armston J, Krishna Moorthy SM, Dubayah R, Costedoat S, Enquist BJ, Fatoyinbo L, Goetz SJ, Gonzalez-Roglich M., Merow, C, Roehrdanz, P. R., Tabor, K. & A. Zvoleff (2023) [The effectiveness of global protected areas for climate change mitigation](#). *Nature Communications*. 2023 Jun 1;14(1):2908.
51. Munoz, F. et al. (2023) [The ecological causes of functional distinctiveness in communities](#). *Letters* 26.8 (2023): 1452-1465.

52. Xu, W. B., Guo, W. Y., Serra-Diaz, J. M., Schrodtt, F., Eiserhardt, W. L., Enquist, B. J. et al. & Ordonez, A. (2023). [Global beta-diversity of angiosperm trees is shaped by Quaternary climate change](#). *Science Advances*, 9(14), eadd8553.
53. Hordijk, Iris, et al. (2023) [Evenness mediates the global relationship between forest productivity and richness](#). *Journal of Ecology* 111(6), 1308-1326.
54. Jónsdóttir, Ingibjörg S., et al. "[Intraspecific trait variability is a key feature underlying high Arctic plant community resistance to climate warming](#)." *Ecological Monographs* (2022): e1555.
55. Keller, A., Ankenbrand, M. J., Bruelheide, H., Dekeyzer, S., Enquist, B. J., Erfanian, M. B., ... & Penone, C. (2023). [Ten \(mostly\) simple rules to future proof trait data in ecological and evolutionary sciences](#). *Methods in Ecology and Evolution* 14(2), pp.444-458.
56. Merow, C., Boyle, B., Enquist, B.J., Feng, X., Kass, J.M., Maitner, B.S., McGill, B., Owens, H., Park, D.S., Paz, A. and G.E. Pinilla-Buitrago (2023) [Better incentives are needed to reward academic software development](#). *Nature Ecology & Evolution*, pp.1-2.

2022

57. Chacón-Labela, J., Hinojo-Hinojo, C., Bohner, T., Castorena, M., Violle, C., Vandvik, V. and B.J. Enquist (2022) [How to improve scaling from traits to ecosystem processes](#). *Trends in Ecology & Evolution*, 38(3), pp:228-237.
58. Castorena, M., Olson, M.E., Enquist, B.J. and A. Fajardo (2022) [Toward a general theory of plant carbon economics](#). *Trends in Ecology & Evolution* 37(10), pp: 829-837
59. Pillet, M., Goettsch, B., Merow, C., Maitner, B., Feng, X., Roehrdanz, P., and B.J. Enquist (2022). [Elevated extinction risk of cacti under climate change](#). *Nature Plants* 8, 366–372.
60. Burger, J. R., Okie, J. G., Hatton, I. A., Weinberger, V. P., Shrestha, M., Liedtke, K. J., ... & Enquist, B. J. (2022). [Global city densities: re-examining urban scaling theory](#). *Frontiers in Conservation Science*, 3, 125.
61. Díaz, Sandra, et al. (2022) [The global spectrum of plant form and function: enhanced species-level trait dataset](#). *Scientific Data* 9.1: 755.
62. Zuidema, P. A., Babst, F., Groenendijk, P., Trouet, V., Abiyu, A., Acuña-Soto, R., et al. & Zhou, Z. K. (2022). [Tropical tree growth driven by dry-season climate variability](#). *Nature Geoscience*, 1-8.
63. Guo, Wen-Yong, et al. (2022) [High exposure of global tree diversity to human pressure](#). *Proceedings of the National Academy of Sciences* 119.25: e2026733119.
64. Krieger, J., Duarte, G., Boakes, E., Capellao, R., Chaplin-Kramer, R., Enquist, B., Feng, X., Gomes, F., et al. (2022) [Ambitious goals require inclusive and integrated spatial planning: Recommendations for the post-2020 Global Biodiversity Framework \(2022\)](#) International Institute for Sustainability. <http://pure.iiasa.ac.at/id/eprint/17841/>
65. Chaplin-Kramer, R., Brauman, K. A., Cavender-Bares, J., Díaz, S., Duarte, G. T., Enquist, B. J., ... & Zafra-Calvo, N. (2022). [Conservation needs to integrate knowledge across scales](#). *Nature Ecology & Evolution*, 6(2), 118-119.
66. Feng, X., Enquist, B. J., Park, D. S., Boyle, B., Breshears, D. D., Gallagher, R. V., ... & López-Hoffman, L. (2022) [A review of the heterogeneous landscape of biodiversity databases: opportunities and challenges for a synthesized biodiversity knowledge base](#). *Global Ecology and Biogeography*. 2022 Apr 18.

67. Boyle, B. L., Maitner, B. S., Barbosa, G. G., Sajja, R. K., Feng, X., Merow, C., ... & Enquist, B. J. (2022). [Geographic name resolution service: A tool for the standardization and indexing of world political division names, with applications to species distribution modeling](#). *Plos One*, 17(11), e0268162.
68. Maitner, B. S., Park, D. S., Enquist, B. J., & Dlugosch, K. M. (2022). [Where we've been and where we're going: the importance of source communities in predicting establishment success from phylogenetic relationships](#). *Ecography*, 2022(6), e05406.
69. Aguirre-Gutiérrez, Jesús, et al. [Functional susceptibility of tropical forests to climate change](#). *Nature Ecology & Evolution* (2022): 1-12.
70. Wieczynski, D.J., Díaz, S., Durán, S.M., Fyllas, N.M., Salinas, N., Martin, R.E., Shenkin, A., Silman, M.R., Asner, G.P., Bentley, L.P., Malhi, Y., B.J. Enquist and V.M. Savage (2022) [Improving landscape scale productivity estimates by integrating trait-based models and remotely-sensed foliar trait and canopy structural data](#). *Ecography*, p.e06078.
71. Lourenço Junior J, Enquist BJ, von Arx G, Sonsin Oliveira J, Morino K, Thomaz LD, Milanez CR. (2022) [Hydraulic tradeoffs underlie local variation in tropical forest functional diversity and sensitivity to drought](#). *New Phytologist*. 234 (1), 50-63
72. Gatti, R.G. et al. (2022) [The number of tree species on Earth](#). Proceedings of the National Academy of Sciences, U.S.A DOI: 10.1073/pnas.2115329119
73. Maitner BS, Park DS, Enquist BJ, Dlugosch KM. (2022) [Both source and recipient range phylogenetic community structure can predict the outcome of avian introductions](#). *Ecography*.
74. Brun, P., Violle, C., Mouillot, D., Mouquet, N., Enquist, B.J., Munoz, F., Münkemüller, T., Ostling, A., Zimmermann, N.E. and W. Thuiller (2022). [Plant community impact on productivity: Trait diversity or key \(stone\) species effects?](#) *Ecology Letters*. 25(4), 913-925.

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- 336. Mitchell, R, D. Bleakly, R. Cabin, R. Chan, B.J. Enquist , A. Evans, T. Lowrey, D. Marshall, S. Reed, G. Stevens, & N. Wasser (1993). [Species concepts, response to Ernst Mayr: Speciation and hybridization in plants](#). *Nature* 364:20.

Other Publications

Electronic Publications and Preprints

- 1- Enquist B.J., Condit R., Peet R.K., Schildhauer M. & Thiers B.M. (2009). The Botanical Information and Ecology Network (BIEN): Cyberinfrastructure for an integrated botanical information network to investigate the ecological impacts of global climate change on plant biodiversity. [The iPlant Collaborative]. URL www.iplantcollaborative.org/sites/default/files/BIEN_White_Paper.pdf

- 2- West, G.B., Savage, V.M., Gillooly, J., Enquist, B.J. Woodruff, W.H. and J.H. Brown (2002). Red herrings and rotten fish. arXiv. <<http://arxiv.org/abs/physics/0211058>>.
- 3- Fuller, M.M., Enquist, B.J. and A. Wagner. Neutrality is Size-Dependent in Tropical Understory Trees. A preprint of this manuscript is available as a *Santa Fe Institute Working Paper* www.santafe.edu/research/publications/wplist/2004
- 4- Fuller, M.M., Enquist, B.J. and A. Wagner. Species Association Networks of Tropical Trees Have a Non-neutral Structure. A preprint of this manuscript is available as a *Santa Fe Institute Working Paper* www.santafe.edu/research/publications/wplist/2004.

Field Guides, Natural History Publications, Botanical Collections

1. Enquist, B. & J. Sullivan (2001) [Vegetative key and descriptions of tree species of the tropical dry forests of upland Sector Santa Rosa](#), Área de Conservación Guanacaste, Guanacaste, Costa Rica. Published electronically via the Área de Conservación Guanacaste.
2. Lucely L. Vilca, Paul Santos, Jonathan Sallo, Vigdis Vandvik, Brian J. Enquist and William Farfan-Rios (2020) [Common Plants of the grasslands of Manu --Manu Biosphere Reserve, Paucartambo, Cusco, Peru](#); 2800 to 4100 msnm. Environmental & Conservation Programs, The Field Museum, Chicago, IL 60605 USA. https://paulefrensa.rbind.io/publication/2022-02-01-manu_grassland/ [rrc@fieldmuseum.org] [www.fmnh.org/plantguides/] <https://fieldguides.fieldmuseum.org/guides/guide/1452>
3. Botanical Collections associated with my laboratory field campaigns are deposited in numerous herbaria. Many are in the University of Arizona Herbarium and can be accessed via SEINet here <https://swbiodiversity.org/seinet/collections/harvestparams.php?db=2> Search for Enquist under 'Collector Criteria'.

Commentary, News, & Popular Literature

- Cotner, S., Enquist, B.J., Farfan-Rios, W., Michaletz, S., Gauthier, T.J., Vandvik, V., Hošková, K., Pierfederici, M.E., Diaz, E.S., Jessup, L., Strydom, T., and Von Oppen, J. (2020) [International scientists need better support during global emergencies. Times Higher Education](#).
- Enquist, B.J. (2023) [Tree Theory, Biogeography, Branching](#). pp 29-33. In: [The Language of Trees: A Rewilding of Literature and Landscape](#) Ed. Katie Holten. Tin House Press. New York, NY.. Over 50 writings from notable authors, philosophers, scientists and artists.
- Enquist, B.J. (2015) [Tree Theory, Biogeography, Branching](#). pp 31-35. In: *All About Trees* Ed. Katie Holten. Dimanche Press. Berlin, Germany. A collection of essays and works. from 50 different authors / scientists / poets

Data and Databases

- **BIEN** - Enquist, B.J. et al. [The Botanical Information and Ecology Network \(BIEN\) dataset: A global compilation of plant observation data and biodiversity informatics workflow](#). <https://biendata.org/> also accessible via, Maitner, B.S., B. Boyle, N. Casler, R.

- Condit, J. Donoghue II, S. M. Durán, D. Guaderrama, C. E Hinchliff, P. M. Jørgensen, N. J. B. Kraft, B. McGill, C. Merow, N. Morueta-Holme, R. K. Peet, B. Sandel, M. Schildhauer, S. A. Smith, Jens-Christian Svenning, B. Thiers, C. Violle, S. Wiser, and B. J. Enquist. (2018) [The BIEN R package: A tool to access the Botanical Information and Ecology Network \(BIEN\) Database](#). *Methods in Ecology and Evolution*, 9:373-379.
- **ForestGEO - San Emilio Forest Dynamics Plot (SEFDP) Over the past ~45 years, field crews have censused the San Emilio plot four times.** The forest plot is 15.64-ha, subject to a distinct wet (June-December) and dry (January-May) season, and located within the Área de Conservación de Guanacaste (ACG) in the Guanacaste Province of Costa Rica. Brian Enquist and Nate Swenson are the Principal Investigators of the San Emilio plot. <https://forestgeo.si.edu/sites/san-emilio>
 - **Global Plant Biomass Allocation and Growth Data.** Niklas, K. J., and B.J. Enquist. 2004. *Biomass Allocation and Growth Data of Seeded Plants. Data set.* Available on-line, <<http://www.daac.ornl.gov>> from Oak Ridge National Laboratory Distributed Active Archive Center, Oak Ridge, Tennessee, U.S.A.
 - **CHAMBASA, Peru Forest Plot Trait Data** - Shenkin, A., Bentley, L. P., Girardin, C., Blonder, B., Boyle, B., Doughty, C. E., ... Malhi, Y. (2017). GEMTraits: A database and R package for accessing and analyzing plant functional traits from the Global Ecosystems Monitoring Network. Oxford University Research Archive. Retrieved from <https://doi.org/10.5287/bodleian:voBDo4N7o>
 - **Peru PFTC Data** - Halbritter, A. H., Vandvik, V., Cotner, S. H., Farfan-Rios, W., Maitner, B. S., Michaletz, S. T., ... & Enquist, B. J. (2024). [Plant trait and vegetation data along a 1314 m elevation gradient with fire history in Puna grasslands, Perú](#). *Scientific Data*, 11(1), 225.
 - **Svalbard PFTC Data** - Vandvik, V., Halbritter, A. H., Althuizen, I. H., Christiansen, C. T., Henn, J. J., Jónsdóttir, I. S., ... & Enquist, B. J. (2023). [Plant traits and associated data from a warming experiment, a seabird colony, and along elevation in Svalbard](#). *Scientific Data*, 10(1), 578.
 - **China PFTC Data** - Vandvik, V., Halbritter, A.H., Yang, Y., He, H., Zhang, L., Brummer, A.B., Klanderud, K., Maitner, B.S., Michaletz, S.T., Sun, X. and Telford, R.J. et al. and B.J. Enquist 2020. [Plant traits and vegetation data from climate warming experiments along an 1100 m elevation gradient in Gongga Mountains, China](#). *Scientific data*, 7(1), pp.1-15.
 - **Norway PFTC Data** - Coming soon!
 - **South Africa PFTC Data** - Coming soon!

Software and Online Biodiversity Informatics Tools

- **TNRS - The Taxonomic Name Resolution Service** – webservice and API - <https://tnrs.biendata.org/> Standardize your plant names **here** using the TNRS web site or **here** using the TNRS R package. The TNRS is a critical tool for integration of any botanical data. Click [here](#) for the TNRS publication. *Nature* featured the TNRS [in a News & Views story](#).
- **GNRS - The Geographic Name Resolution Service** – webservice and API <https://gnrs.biendata.org/> like the TNRS but for geographic names. Standardize your

- geographic names and locations **here**. The GNRS corrects and standardizes political division names according to the standards of the GeoNames database. Standardization of country, state and county names is the crucial first step in validating the accuracy of geocoordinates ("geovalidation"). BIEN geovalidations include checking that the coordinates of an observation fall within the boundaries of its declared political divisions.
- **NSR - Native Species Resolver** – webservice and API <https://nsr.biendata.org/> This specimen record of *Cattleya skinneri* in Peru, is it native or introduced? Ask the NSR! Currently, the NSR can be accessed via a web service and in batch mode (shell access to server required).
 - **GVS - Geocoordinate Validation Service** -webservice and API <https://gvs.biendata.org/> (coming soon!) The Geocoordinate Validation Service (GVS) is a tool for the detection of errors in geographic coordinates.
 - **occTest - An integrated approach for quality control of species occurrence data.** <https://github.com/pepbioalerts/occTest> Species occurrence data are valuable information that enables one to estimate geographical distributions, characterize niches and their evolution, and guide spatial conservation planning. However, persistent quality issues in occurrence data can impact the accuracy of scientific findings. offers a structured set of algorithms to identify potential problems with species occurrence records by employing a hierarchical organization of multiple tests.
 - **R Package** - Maitner, B.S., B. Boyle, N. Casler, R. Condit, J. Donoghue II, S. M. Durán, D. Guaderrama, C. E Hinchliff, P. M. Jørgensen, N. J. B. Kraft, B. McGill, C. Merow, N. Morueta-Holme, R. K. Peet, B. Sandel, M. Schildhauer, S. A. Smith, Jens-Christian Svenning, B. Thiers, C. Violle, S. Wiser, and B. J. Enquist. (2018) [The BIEN R package: A tool to access the Botanical Information and Ecology Network \(BIEN\) Database](#). *Methods in Ecology and Evolution*, 9:373-379.
 - **R Package occTest** - Serra-Diaz JM, Borderieux J, Maitner B, Boonman CC, Park D, Guo WY, Callebaut A, Enquist BJ, Svenning JC, Merow C. [occTest: An integrated approach for quality control of species occurrence data](#). *Global Ecology and Biogeography*. 2024:e13847
 - **R Package RTNRS** - <https://github.com/EnquistLab/RTNRS> The TNRS R package (also known as RTNRS) provides access to the Taxonomic Name Resolution Service API, which is a tool for automated standardization of plant scientific names. The TNRS corrects spelling errors and alternative spellings to a standard list of names, and converts out of date names (synonyms) to the current accepted name.
 - **R Package RGNRS** - <https://github.com/EnquistLab/RGNRS> The GNRS package for R (a.k.a. the RGNRS) provides users with access to the Geographic Name Resolution Service (GNRS) API in R. The Geographic Name Resolution Service takes in the names of political units and standardizes them following <https://www.geonames.org/>
 - **R Package RNSR** - <https://github.com/EnquistLab/RNSR> The NSR package for R (a.k.a. the RNSR) provides users with access to the Native Species Resolver (NSR) API in R. The Native Species Resolver takes taxonomic and political information as input and returns information on whether a given plant species is thought to be native, introduced, etc., as well as the basis for that decision.
 - **R Package RCDS** - <https://github.com/EnquistLab/RCDS> - R package for accessing the Centroid Detection Service.

Sampling of Open-source Lectures (Not Peer Reviewed)

- 1- Enquist, Brian (2015): A quick (and rough) introduction to trait-based ecology. figshare. <http://dx.doi.org/10.6084/m9.figshare.1396511>
- 2- Enquist, Brian (2014): Overview lecture on Macroecology. figshare. <http://dx.doi.org/10.6084/m9.figshare.1247653>
- 3- Enquist, Brian (2014): Introduction to Metabolic Scaling Theory – From cells to ecosystems. figshare. <http://dx.doi.org/10.6084/m9.figshare.1275197>.
- 4- Enquist, Brian (2016): How to think About Your Data: Introduction to Data Science & Management (what they don't – but should – teach you about the scientific method). figshare. <https://dx.doi.org/10.6084/m9.figshare.4251953.v1>
- 5- Enquist, Brian; Savage, Van (2017): Trait driver theory: A basis to integrate and scale from plant form, function and strategies to ecosystems worldwide. figshare. <https://doi.org/10.6084/m9.figshare.5328004.v1>
- 6- Enquist, Brian; Merow, Cory; McGill, Brian; Boyle, Brad; Casler, Nathan; Feng, Xiao; et al. (2018): Muy BIEN: Next steps in a global workflow for integrating plant botanical observations. figshare. Paper. <https://doi.org/10.6084/m9.figshare.6983024.v1>
- 7- Enquist, Brian (2020). #PlantBlindness: Why plants matter and why we study traits.. figshare. Presentation. <https://doi.org/10.6084/m9.figshare.12084966.v1>
- 8- Enquist, Brian (2020). An introduction to trait based ecology. figshare. Presentation. <https://doi.org/10.6084/m9.figshare.11704383.v1>

Grants, Awards, and Fellowships

- **National Science Foundation, BioDiversity on a Changing Planet (2022-2025), *BoCP: BioFI Biodiversity Forecasting Initiative to Understand Population, Community, and Ecosystem Function Under Global Change***. PI Brian Enquist
- **Department of Energy (2020-2023) *Biogeochemical Dynamics from Genomes to Watershed Scales; The Watershed Function Scientific Focus Area (SFA)*** PIs Susan Hubbard and Ken Williams, LBNL, Berkeley Lab.
- **National Science Foundation, Harnessing the Data Revolution, Harnessing the Data Revolution, (2019-2022) *Collaborative Research: Near term forecasts of global plant distribution, community structure, and ecosystem function***. PI B.J. Enquist
- **National Geographic Explorers Grant (2019-2021) *Analyzing the future of tropical dry forests using new technologies and low-cost laser scanning***. PIs B.J. Enquist and Adam Chumurzynski.
- **Department of Energy (2017-2020) *Biogeochemical Dynamics from Genomes to Watershed Scales; The Watershed Function Scientific Focus Area (SFA)*** PIs Susan Hubbard and Ken Williams, LBNL, Berkeley Lab.
- **Norwegian Research Council (2017 - 2023) *TraitTrain: Comparing climate change impacts on High North vs. Alpine ecosystems through research and training in trait-based approaches***. PI Vigdis Vandvik;
- **Norwegian Research Council (2018 - 2021) *INCLINE: Indirect climate change***

- impacts on alpine plant communities (Forskerprosjekt - FRIMEDBIO).* PI Vigdis Vandvik.
- **National Science Foundation (2016-2019)** *ABI Development: Creating a generic workflow for scaling up the production of species ranges* PI B.J. Enquist; Collaborative proposal with Brian McGill (University of Maine) and Cory Merrow (Yale University).
 - **National Science Foundation (2016-2019)** *RUI: Niche evolution, ecological limits, and the macroecology of land plant biodiversity.* NSF (DEB Populations and Community Ecology). PI B.J. Enquist; Collaborative proposal with A. Kerkhoff (Kenyon College).
 - **National Science Foundation (2015-2018)** *Developing integrated trait-based scaling theory to predict community change and forest function in light of global change.* NSF (DEB Ecosystems). PI B.J. Enquist; Collaborative proposal with V.M. Savage (UCLA), L.Patrick-Bentley, and G. Asner (Carnegie/Stanford).
 - **Global Environment Facility (GEF) and Conservation International (2016-2019)** *Spatial Planning for Protected Areas in Response to Climate Change (SPARC)* PI Lee Hannah.
 - **Department of Homeland Security (2016-2018).** *A general informatics workflow, database, and geospatial tools for the dissemination of high quality biodiversity data across space and time.* PI B.J. Enquist.
 - **NERC: UK Research Grant (2013 – 2015)** - *Tree communities, airborne remote sensing and ecosystem function: new connections through a traits framework applied to a tropical elevation gradient.* PI Yadvinder Malhi (Oxford University).
 - **University of Arizona, Institute for the Environment (2013)** - *Process-based Models for Forecasting Geographic Ranges – Integrating Plant Distributions, Demography, and Traits in a Bayesian, Informatics Framework.* Institute for the Environment Faculty Exploratory Grant, PIs B.J. Enquist and M. Evans.
 - **Aspen Center for Environmental Studies (ACES) Grant (2013 – 2015)** - *Future Forest Distribution Model” and website in order to forecast and visualize Western tree species responses to climate change.*
 - **National Science Foundation – Dissertation Improvement Grant (2008 – 2010)** Graduate fellowship for Benjamin Blonder – *Paleoclimate reconstruction from leaf venation networks*, NSF DEB.
 - **National Science Foundation (2013-2016).** *Novel lessons from ancient plants: Water transport in the earliest tracheophytes.* Jarmilla Pitterman, PI, UC Santa Cruz.
 - **National Science Foundation (2012-2015).** Research Opportunity Award (ROA). PI Enquist, B.J., Supplemental funding to - *Collaborative Research: Experimental Macroecology: Effects of Temperature on Biodiversity.*
 - **National Science Foundation, Macrosystems (2011- 2015).** *Experimental Macroecology.* Collaborative proposal with J. H. Brown, Kaspari, M., Wade, B., J. Zhou, C. Hou.
 - **National Science Foundation via the iPlant Collaborative (2010 – 2012).** *Developing BIEN – The Botanical Information and Ecology Network - to investigate the ecological impacts of global climate change on plant biodiversity.* PI B.J. Enquist, Co-PI Condit, R., Theirs, B., M. Schildhauer and B. Peet.
 - **National Science Foundation (2010 – 2013).** *Semantic Web Informatics for Species in Space and Time.* PI Dunne, J., Co-PI B.J. Enquist, J. Golbeck, R. Williams and N. Martinez.

- **National Science Foundation (2008-2012).** Major Research Instrumentation. Acquisition of a Distributed Environmental Sensor Network by the Rocky Mountain Biological Laboratory PI: Ian Billick, Co-PIs: C.Still, B.J. Enquist, B. Peckarsky, and S. Wissinger.
- **National Center for Ecological Analysis and Synthesis and iPlant (2008 – 2012).** **Enquist, B.J.** PI with Co-PIs R. Condit, R. Peet, B. Boyle, and S. Dolins, National Center for Ecological Analysis and Synthesis (NCEAS) working group, “Developing BIEN – The Botanical Information and Ecology Network - to investigate the ecological impacts of global climate change on plant biodiversity”. Support of participant costs for 5 working group meetings 2008 - 2012.
- **National Science Foundation – Advancing Theory in Biology, ATB (2008-2011)** Combining Theories For Plant Architecture, Allometry, and Traits to Develop the Next Generation of Scaling Theory **Enquist, B.J.** PI, Collaborative proposal with V. Savage (UCLA), J. Sperry (Utah), and P. Reich (Minnesota).
- **National Science Foundation – Dissertation Improvement Grant (2008 – 2010).** Graduate Fellowship for Scott Stark – Testing multiple determinants of growth rate-mass scaling relationships in an Amazonian Forest.
- **Center for Applied Biodiversity Science, Conservation International, CABS Fellowship (2008-2009).** Developing a general macroecological framework for scaling plant community, diversity, biomass, and dynamics across time and space.
- **National Park Service Research Grant (2008-2009)** Assessing plant taxonomic and functional diversity across elevational gradients. Enquist, B.J. PI, B.Boyle (Co-PI).
- **Center for Applied Biodiversity Science, Conservation International, CABS Fellowship (2007-2008).** Developing a general macroecological framework for scaling plant community, diversity, biomass, and dynamics across time and space. PI: B.J. Enquist.
- **National Science Foundation Post-doctoral BioInformatics Fellowship (2006-2008)** (E.P. White (PI), Enquist, B.J. CoPI).
- **National Science Foundation Young Investigator CAREER Award (2002-2007).** Scaling plant life-history, ontogeny, diversity, and ecology: Elaboration of a general model. PI- B.J. Enquist. NSF DEB-0133974.
- **United States Geological Survey (USGS) Post-doctoral Research Grant (2003-2004)** Developing a general framework for scaling plant community diversity, biomass, and dynamics across time and space – SALVIAS (Synthesis and Analysis of Local Vegetation Inventories Across Scales). PI - Enquist, B.J., Co-PI J. Pither.
- **National Parks Service Research Grant (2003-2004).** Assessing Floristic Communities along an Elevational Gradient In Tropical Mexico – Comparison to Florissant National Monument, CO.PI Enquist, B.J. Co-PI B. Boyle and D. Kerkhoff.
- **Center for Applied Biodiversity Science, Conservation International, CABS Fellowship (2002-2004)** Developing a general macroecological framework for scaling plant community, diversity, biomass, and dynamics across time and space; PI - Enquist, B.J.
- **Department of Energy (DOE), Los Alamos National Labs (2003-2004)** Scaling Relationships in Biology: Developing and Applying a Unifying Theory from Molecular through Biosphere Scales. PI D. D. Breshears,; Co-PIs Unkefer, P.J., West, G.B., Woodruff, W.H., Donohoe, R.J., Ebinger, M.H., Cremers, D.A.; Collaborators **Enquist, B.J.**, Brown, J.H. and Allen, C.D.

Organized Meetings, Workshops, Symposia

- **2022 International Workshop: Synthesizing Biological Scaling: Towards a Universal Theory** November 15, 2022 – November 17, 2022 Our proposed symposium will bring a much needed and belated round of synthesis in biological scaling. We will bring together aspects of the theory and communicate the syntheses that exist to the broader community.
<https://www.santafe.edu/events/synthesizing-biological-scaling-towards-universal-theory>
- **2018 International workshop: OpenTraits, New Orleans, August, 4-5th, 2018-** Organized and led the first OpenTraits workshop part of a coordinated, international series of meetings focused on facilitating open collaboration and standardization in the collection and sharing of trait data. <http://opentraits.org/>
- **2018-Symposium: Ecoinformatics Advances: Building Technosocial Systems for Open Data and Big Science** Ecological Society of America, Annual Meeting 2018, New Orleans. Organizers: John (Jack) Williams Corina Gries, and B.J. Enquist <https://eco.confex.com/eco/2018/meetingapp.cgi/Session/14041>
- **2018- International Course - International Plant Functional Traits Course 4, Svalbard.** University Centre in Svalbard, Norway. Co-led and co-taught with Prof. V. Vandvik for 20 students
<https://www.uib.no/en/rg/EECRG/114808/plant-functional-traits-course-4>
- **2017- International Workshop: A roadmap for global synthesis of the plant tree of life.** An international workshop held at Kew Royal Botanical Gardens, London UK. October 2017. Organizers W. Eiserhardt, B. Maitner, B.J. Enquist, W. Baker.
- **2017- International Conference: Co-Organizer International Biogeography Society Meetings, Tucson Arizona.**
- **2017- Workshop: Integrating and cleaning biodiversity data: Workflows to model ranges and merge associated ecological, phylogenetic, and trait information.** Organizers: Cory Merow, B.J. Enquist et. al. The 8th Biennial Conference of the International Biogeography Society was held at the University of Arizona, Tucson Arizona, January 2017.
- **2017- International Course - International Plant Functional Traits Course 3, Puna Alpine Grasslands, Peru.** Wayqecha Cloud Forest Biological Station. Co-led and co-taught with Prof. V. Vandvik for 20 students
<https://www.uib.no/en/rg/EECRG/109375/plant-functional-traits-course-3>
- **2016- International Course - International Plant Functional Traits Course 2, Gongga Mountains, Tibetan Plateau, China.** Alpine Ecosystem Observation and Experiment Station. Co-led and co-taught with Prof. V. Vandvik for 20 students
<https://www.uib.no/en/rg/EECRG/97477/plant-functional-traits-course-2>
- **2015-Symposium: Scaling in Ecology: Building a synthetic and predictive science for the next 100 years.** Organized symposium for the 100th Anniversary of the Ecological Society of America meetings, 2015, Baltimore. Organizers: B.J. Enquist and Charles Price. <https://eco.confex.com/eco/2015/webprogram/Session10682.html>
- **2014-International Issue: Proceedings of the National Academy of Sciences, Special Issue on ‘Functional Biogeography’, 2014.** In 2011, led by Cyrille Violle, we proposed to the editors of PNAS to organize a special issue (an ‘in print’

- symposium) on 'Functional Biogeography'. This has been tentatively accepted and the due date for the collection of papers is June 15th 2014. NAS member Prof. Steve Pacala at Princeton University and Peter Reich from University of Minnesota agreed to sponsor this issue. The editors of the session are post-doc Cyrille Violle (CNRS France & Univ. of Arizona), Brian J. Enquist (Univ. Arizona), and Stephen Pacala (NAS member; Princeton Univ.). The Associate Editors: Jens Kattge (Max Planck Institute), Peter Reich (Univ. Minnesota).
- **2012-2008-NCEAS working group, 2012 - 2008** “Developing an integrated botanical information network to investigate the ecological impacts of global climate change on plant biodiversity” – Organized at the National Center for Ecological Analysis and Synthesis or the Botanical Information and Ecology Network (BIEN). PI, B.J. Enquist, Co-PIs Richard Condit, Bob Peet, Brad Boyle, Steve Dolins. October, 2011 <http://www.nceas.ucsb.edu/featured/enquist>
 - **2010-Symposium: Developing a Taxonomic Name Resolution Service of Botany, April 2010.** Working group at the Missouri Botanical Garden, March, 2010. Via the iPlant Collaborative The BIEN team is working to assemble a demonstration project that includes most of the premier plant biodiversity databases for the Americas (sources listed below). By the end of 2011 we propose to produce a single resource giving species names, locations, and often abundances, for about 25 million species occurrence records. Our ultimate goal is to unite an ever-growing pool of plant distributional data with information on plant co-occurrence, ecology, traits and phylogeny. See <http://www.iplantcollaborative.org/node/1044> and <https://pods.iplantcollaborative.org/wiki/display/iptol/TNRS+Workshop>
 - **2010-International Conference, Elected Chair – 2010 Gordon Research Conference on “The metabolic basis of ecology”** held July, 2010. Elected by the participants of the 2006 Gordon Conference to chair the July, 2010 meeting. Together with my co-chair Dr. David Atkinson of Liverpool University, we initiated, organized, advertised, and oversaw the 2010 GRC conference. This was a two year commitment to this GRC meeting that included handing the \$20K GRC budget and fundraising an additional \$9K to support speaker and participant travel and registration costs. <http://www.grc.org/programs.aspx?year=2010&program=metbasis>
 - **2006-Workshop Toward a Unified Theory of Ecological Systems** (J. Green, B.J. Enquist, S. Hubbell, and P. Marquet organizers). Sept. 2006 The Santa Fe Institute and the National Center for Ecological Analysis and Synthesis.
 - **Symposium: A “new” paradigm for community ecology: building from functional ecology** - (McGill, B. and B.J. Enquist organizers). Aug. 2004; Ecological Society of America Meetings in Portland, OR. Program: <http://abstracts.co.allenpress.com/pweb/esa2004/category/?ID=32244>
 - **Symposium: Biophysical Ecology** (Sabo, J. and B.J. Enquist organizers). Aug. 2003; Ecological Society of America Meetings in Savanna GA. Program: <http://abstracts.co.allenpress.com/pweb/esa2003/category/?ID=23502>
 - **Annual Southwestern Association of Biologists Meeting:** Portal, AZ Oct. 2002 (Enquist, B.J. and M. Weiser organizers). Meeting website and official program: <http://eeb37.biosci.arizona.edu/~swab/main.html>.
 - **Symposium: Scaling the implications of organismal size across evolutionary time and ecological space: A synthesis of recent advances and insights.** (Enquist, B.J., Smith, F.A., and P.A. Marquet organizers). Aug 2002;

Ecological Society of America Meetings in Tucson, AZ. Program:
<http://abstracts.co.allenpress.com/pweb/esa2002/category/?ID=510>.

- ***Workshop: Fractals in Biology: Developing the underlying mechanistic principles for self-similarity.*** Workshop with 15 participants, Nov. 29-Dec. 2 2000, The Santa Fe Institute (**B.J. Enquist** and D. Morse organizers). Program:
<http://discuss.santafe.edu/biofractals/>.

Sampling of Public Talks

- 2020 College of Science Public Science Lecture Series, Catalysts of Change. Our Rapidly Changing Biosphere. <https://www.youtube.com/watch?v=GEuvFfl3ZtY>
- 2017 Oxford Martin School – Oxford University. The concept of time in biology, and the unity of life. <https://www.youtube.com/watch?v=u1NJtWQSHQ>
- 2016 UofA Science Café, It's About Time. The Dimensions of Biological Time
- 2015 Life In the Universe – UofA College of Science Public Lecture series. [*Life on Earth: By Chance or By Law?*](#)
- 2014 Aspen Ideas Festival – [*Climate change and the fate of our forests*](#) – Aspen Colorado. July 1st.
- 2012 BrownFest– a homage to Jim Brown and Astrid Kodric-Brown - [*Experimental Allometry & Evolution of Allometry*](#) – Homage to James H. Brown

Supervision and Mentoring

Post-doctoral Researchers/Fellows (20):

Dr. Andrew Kerkhoff (Phd U New Mexico), 2002-2005. Currently Provost, University of Puget Sound; **Dr. Brad Boyle** (PhD Washington U), 2002-2006, 2010-2013; **Dr. Jason Pither** (PhD Queens U), 2004-2007, Associate Professor, Biology Department at the University of British Columbia, Okanagan; **Dr. Ethan White** (PhD U. New Mexico), 2005- 2007, Associate Professor, University of Florida; **Dr. Lisa Patrick** (PhD Texas Tech), Sept. 2008 – 2013. Associate Professor, Sonoma State University; **Dr. Cyrille Violle** (PhD Montpellier University, France), 2009 – 2012. He was a Marie Currie Fellow at the University of Arizona and at the CEFÉ at the University of Montpellier, France. CNRS Scientist (Centre National de la Recherche Scientifique), University of Montpellier, France; **Dr. Sean Michaletz** (PhD University of Calgary), 2013 - 2017. Currently, Assistant Professor, University of British Columbia, Canada; **Dr. Naia Morueta-Holme** (PhD Biology, Aarhus University 2014), 2015 - 2017 Currently, Associate Professor Center for Macroecology, Evolution and Climate, University of Copenhagen, Denmark; **Dr. Sandra Duran** (PhD University of Alberta). 2016 - 2020, Assistant Professor Colorado State University; **Dr. Noah Charney** 2015-2017; Assistant Professor University of Maine Orono; **Dr. Danilo Neves** (PhD Federal University of Minas Gerais, Botany Department, Brazil). 2016 - 2018 , Assistant Professor position at the Federal University of Minas Gerais (Brazil); **Dr. Daniel Park** (PhD Harvard, 2016) BBCS postdoc, 2017-2018, Assistant Professor Purdue University ; **Dr. Xiao Feng** (PhD University of

Oklahoma) BBCS postdoc 2019-2020 Assistant Professor, University of North Carolina ; **Dr. Rachael Gallagher** (PhD Macquarie University, Australia) 2017. She was a ARC Discovery Early Research Career spending part of her postdoc in my lab; **Dr. Robbie Burger** (PhD University of New Mexico), currently Assistant Professor, University of Kentucky; **Dr. Erica Newman** (PhD UC Berkeley, 2016) 2018 - 2020; **Dr. Julia Chacón-Labella** (PhD Madrid 2018) 2020 - 2022, Assistant Professor ; **Dr. Teresa Bohner** (PhD UC Riverside) USGS Scientist; **Dr. Cesar Hinojo Hinojo** (PhD UC Irvine) 2021-2023 Assistant Professor University of Sonora, Mexico; **Dr. Gabriel Moulatlet** (2024-present)

Graduate students, PhD (20):

Charles Price (PhD EEB: 2001 – 2006), Associate Professor position at the University of Western Australia; **Michael Weiser** (PhD EEB: 2001 – 2007); **Megan McCarthy** (PhD EEB:2002 – 2006); **Nathan Swenson** (PhD EEB:2003 – 2008); **James Stegen** (PhD:2004 – 2009); **Christine Lamanna** (PhD EEB:2006 –2012) climate-change ecologist and decision analyst with ICRAF's Climate Change Unit, The Center for International Forestry Research and World Agroforestry; **Scott Stark** (PhD EEB:2006 – 2012) Associate Professor, Michigan State University; **Cathy Hulshof** (PhD EEB:2007 – 2012). **Lindsey Sloat** (PhD EEB:2009 – 2015) Land and Carbon Lab, World Resources Institute; **Benjamin Blonder** (PhD EEB:2009 – 2014). Assistant Professor University of California, Berkeley; **Julie Messier** (PhD EEB:2010 – 2015) Assistant Professor, University of Waterloo, Canada; **Alex Brummer** (PhD Physics:2012 – 2017). Department of Physics and Astronomy, College of Charleston; **Vanessa Buzzard** (PhD EEB:2013 – 2017). Research Scientist Biosphere2; **Brian Maitner** (PhD EEB:2014 – 2021) Assistant Professor University of South Florida; **Lorah Patterson** (PhD EEB:2016 – 2025); **Adam Chmurzynski** (PhD EEB:2017 – 2025); **Michiel Pillet** (PhD EEB:2018 – 2025); **Jocelyn Navarro** (PhD EEB:2020 -); **Connor Wilson** (PhD EEB:2022 -) ; **Michael Mustri** (PhD EEB: 2023-); **Maryela Celaya** (2024 - present);

Graduate students, Masters (4):

John Donoghue (Masters:2010 – 2014). Co-advised with B. McGill, GIS Manager, GIS Engineer, The Altum Group; **Amanda Henderson** (Masters:2013 – 2016). DOE Research Scientist, Berkeley Labs; **Matiss Castorena Salaks**, Masters: 2017 – 2022);

Undergraduate students

Numerous undergraduates have worked in my lab assessing with plant trait and stoichiometry measures as well as helping in the field and with various lab research. Too many to count. Many have done honors theses and completed research projects. Some notable undergraduates who have completed research study include: **Evan Economo** - a faculty member at the Okinawa Institute of Science and Technology where I lead the Biodiversity and Biocomplexity Unit

<https://groups.oist.jp/bbu/evan-p-economo>; **John Grady** Postdoctoral Fellow, <https://www.johnmgrady.com/> **Rachael Mitchel** Assistant Professor – University of Arizona – School of Natural Resources and the Environment. <https://mitchell-ecology.com/>; **Peter Gaube** Senior Principal Oceanographer, Applied Physics Laboratory, University of Washington <https://submesoscale.org/>

International graduate students and post-docs (8):

Hélène Morlon (2011; **Kristine Engemann Jensen** (2012 – 2015). She finished her PhD with Jens-Christian Svenning at Aarhus University, Denmark. I was an international member of her PhD committee and hosted her for in my lab for several months in 2012-2015; **Mia Vedel Sørensen** (2016 – 2018) I was co-supervisor with Professor Bente Jessen Graae at the Norwegian University of Science and Technology in Trondheim Norway (NTNU). Mia is applying our methods for linking traits and ecosystem carbon flux in a long-term experiment along an elevational gradient in Norway; **Dr. Dr. Irena Simova** (PhD Charles University, Czech Republic), August 2012 - 2017. Dr. Simova was an international postdoctoral fellow with funding from the Czech Republic to travel to my lab part of her time and to collaborate on our NSF Macrosystems project; **Jehová Lourenço Junior** – External, PhD Student, Brazil, March 2017 – October 2018, PhD ‘Sandwich’. Jehová defended his PhD at the University of Arizona Fall 2018. He is currently a postdoc Jitka Klimešová in the Czech Republic. **Yaoqi Li** – Visiting PhD Student, Peking University, China, Nov. 2017 – Nov. 2018. Yaoqi is defending her PhD late spring 2019; **Jiaxin Zhang** – Visiting PhD Student, Chinese Academy of Science, Wuhan China, June 2018 – June 2019; **Ragnhild Gya** PhD Student, University of Bergen, Department of Biology, Norway, Sept. 2019 – Sept. 2020.

Affiliations

American Association for the Advancement of Science, American Society of Naturalists, Ecological Society of America (*Theoretical and Physiological Ecology Sections*), Botanical Society of America (*Ecological and Tropical Biology section*).

Overview of synergistic activities

Development of Botanical Ecoinformatics Data Networks, Workflows, and Tools -

In order to enable the next generation of big-data driven science, we have actively worked to develop publically accessible standardized botanical datasets, workflows, and informatics tools. This work includes development of the following:

- **BIEN**- The Botanical Information and Ecology Network – is integrating the available herbarium specimens, plant functional trait and ecological plot data for the New World, biendata.org. BIEN is a network of numerous scientists from across the globe who are developing a standardized workflow to integrate disparate sources of botanical data that together can address prominent questions of plant macroecology, evolution, and comparative biology. The BIEN database currently houses about 200 million integrated and standardized botanical observations from across the New World – from herbarium,

- trait, and ecological plot observations.
- **TNRS** - In order to do repeatable, robust global ecology and comparative biology one must integrate disparate data containing differing taxonomic names. As part of my research efforts to integrate global data collaborative research with the iPlant Collaborative and the Missouri Botanical Garden, we have just developed a Taxonomic Name Resolution Service (TNRS). Featured in *Nature*, the service standardizes and ‘scrubs’ global botanical names so as to make taxon names ‘usable’ for comparative analyses. It is an informatics tool to correct and align any botanical taxonomic information, tnrs.iplantcollaborative.org/ Since its release in 2011, use of the TNRS has been exponential with over 50,000 users and 1,000 new sessions each month.
 - **The San Emilio Forest Dynamics long-term plot** - My group maintains a long-term forest dynamics plot on the growth, death, and recruitment of all trees within a 17ha plot in Guanacaste, Costa Rica. Started in 1976, the plot contains ~ 20,000 trees and is one of the largest and longest running forest plots.
 - **OpenTraits.org** Functional traits have rapidly become an integral part of biodiversity, earth system, and climate change research. Trait-based approaches allow comparisons (of patterns, trends, global change impacts etc.) across regions and systems that have few taxa in common. But despite widespread interest, we still have incomplete knowledge of how traits vary within and across species, locations and time and we still lack a common trait standard to link and share trait data within and across disparate organisms (plants, insects, birds, mammals etc.). I have led the start of OpenTraits to help advance the use of trait based approaches and data in biology and earth system research.

Highlights of some of my lab groups popular science, education & outreach activities

– My lab group has worked with multiple media outlets to teach and promote the broader impact of Science done in the lab. These efforts utilize innovative technological and pedagogical approaches to convey the unique mathematical and large-scale ecological approaches that we use in my lab. This work has included:

- **ForestForecasts** - at forestforecasts.org We have developed in partner with the Aspen Center Environmental Studies a web visualization portal to show the impacts of differing climate change scenarios on local forests in Colorado. We are actively working to design this exploratory visualization tool to explore and understand how different climate change scenarios will impact western forests. I am the PI of this outreach effort to develop novel web accessible visualization and simulation tools to allow students, the public, non-scientists, and scientists to visualize the change of forests with different climate change scenarios. The site, includes links for elementary school education lessons as well as for the general public.
- **Plant-O-Matic** - In order to deliver biodiversity data to the people, we have developed a novel app (freely available on the iTunes store) that uses 100,000 species range maps in the New World let the user know what plants may immediately surround them in their current location. The app provides a fast conduit and rapid access to biodiversity data and is a creative tool by which anyone with a smart phone can interact and discover biodiversity, <https://itunes.apple.com/us/app/plant-o-matic/id906932765?mt=8>
- **PBS NOVA documentary**- “Hunting the Hidden Dimension” focused on the history and study of self-similarity in nature. Filming this documentary featured my work at the Santa Fe Institute as well as within the Area de Conservación Guanacaste, Costa Rica. My work on the film consisted of much logistical planning and assisting with animations and editing script. The documentary aired in 2009 and has now been shown throughout the

world including the public broadcasting channels in most of Europe and Australia. The film focuses on how my work on 'self-similarity' applies to pressing issues in biology, ecology, and global change, see

<http://www.pbs.org/wgbh/nova/physics/hunting-hidden-dimension.html> ; watch us watch starting at ~ 43:00 <https://www.youtube.com/watch?v=HvXbQb57lsE>

- National Geographic Channel, documentary, X-Ray Earth - We worked with the developers to film several sessions at the Rocky Mountain Biological Lab. The series aired in 2011 and focus' on our work in using metabolic scaling theory and new technology to predict and understand the flux of CO₂ from terrestrial ecosystems, see <http://www.natgeotv.com/int/x-ray-earth> ; watch us starting at ~ 50:00 <https://www.youtube.com/watch?v=HYdeYuk3Qa4>
- Popular Science Book – “In the Beat of a Heart” My lab is featured prominently in this popular science book. We conducted several interviews, and hosted the author for fieldwork and multiple excursions. It features my collaborative research on scaling and teaches the linkages between math, large scale ecology, biodiversity, and our work to understand and conserve biodiversity and tropical forests. <https://www.theguardian.com/science/2015/feb/26/in-the-beat-of-a-heart-by-john-whitfield-review>